9 Databases

Cambridge IGCSE Computer Science (2210) — revision notes

In this chapter: single-table databases · fields, records & validation · the 6 basic data types · primary keys · SQL (SELECT, FROM, WHERE, ORDER BY, SUM, COUNT).

9.1.1 Single-Table Databases

Database: a structured collection of data allowing people to extract information to meet their needs. A single-table database contains only one table (relational/multi-table DBs are studied at A-Level).

Why are databases useful?

- Changes/additions only need to be made once — data stays consistent
- The same data is available to everyone who needs it
- In relational databases, data is stored once only — no duplication

Used to store information about people (patients, pupils), things (cars, library books) or events (bookings, race results).

Fields & records — the building blocks

Table: data about ONE type of item/person/event, made up of many records.

Record: a ROW in the table — all data about ONE item/person/event.

Field: a COLUMN in the table — one specific piece of information, given a meaningful name (no spaces), e.g. `FirstName`, `DateOfAdmission`.

Remember: every record in a table has the SAME number of fields — but the NUMBER of records can grow or shrink as records are added/deleted.

9.1.2 The Six Basic Data Types

Syllabus data type	Description	MS Access equivalent
text / alphanumeric	a number of characters	Short Text / Long Text
character	a single character	Short Text, field size = 1
Boolean	True/False, 1/0, Yes/No	Yes/No
integer	whole number	Number, 0 decimal places
real	decimal number	Number, decimal format
date/time	date and/or time	Date/Time

Validation in a database: some checks are automatic (e.g. a Date/Time field rejects "99" automatically); others must be SET UP by the developer — e.g. a validation rule `>=1 And <=10` with validation text "Ward numbers are 1 to 10".

9.1.3 Primary Keys

Primary key: a field whose value is UNIQUE for every record — used to reliably identify one specific record.

- Can be an existing field IF it is guaranteed unique (e.g. ISBN in a book table)
- Otherwise, add a NEW field purely to act as a unique identifier, e.g. `HospitalNumber` in a patient table

Common mistake: choosing FirstName or FamilyName as a primary key — names are often repeated between different people, so they are NOT suitable.

9.1.4 SQL – Structured Query Language

SQL (pronounced "es-queue-el") is used to write scripts that extract exactly the information needed from a database.

Statement	Purpose
SELECT	choose which fields (columns) to display – mandatory, always first
FROM	state which table to use – mandatory
WHERE	filter records (rows) matching a condition – optional
ORDER BY	sort the results (add DESC for descending) – optional
SUM(Field)	total of a numeric field, used with SELECT
COUNT(Field)	counts how many records match a condition, used with SELECT

An SQL command always starts with SELECT and ends with a semicolon ;

```
SELECT Field1, Field2       -- or SELECT * for all fields
FROM TableName
WHERE Condition
ORDER BY Field1;
```

Formatting values in a WHERE condition

Data type	How to write the value
text / char	single quotes: `'Mr Smith'`
Boolean	TRUE or FALSE
integer / real	no quotes: `12` or `12.01`
date/time	single quotes (Access: hashes #): `'22/11/2022'`

Operators used in WHERE conditions

=	equal to	>	greater than	<	less than

>=	≥	<=	≤	<>	not equal to
BETWEEN	range of 2 values	LIKE	pattern match	IN	list of values
AND	all conditions true	OR	at least one true	NOT	condition is false

Worked examples (single-table PATIENT database)

List Mr Smith's patients:

```
SELECT HospitalNumber, FirstName, FamilyName
FROM PATIENT
WHERE Consultant = 'Mr Smith';
```

Same, but in alphabetical order of family name:

```
SELECT HospitalNumber, FirstName, FamilyName
FROM PATIENT
WHERE Consultant = 'Mr Smith'
ORDER BY FamilyName;
```

Count patients in Ward 7:

```
SELECT COUNT(WardNumber)
FROM PATIENT
WHERE WardNumber = 7;
```

Patients admitted between two dates:

```
SELECT FirstName, FamilyName
FROM PATIENT
WHERE DateOfAdmission BETWEEN '12/10/2022' AND '30/10/2022';
```

Exam tips:

- SELECT * means "all fields" — don't list them individually if * is used
- WHERE always comes straight after FROM; ORDER BY always comes LAST
- Text/date values need quotes; numbers do not

- Statement ends with a semicolon `;`

Practical Database Skills (for coursework/ practicals)

As an IGCSE student you should be able to:

1. Define a single-table database structure from a given data requirement (choose field names + suitable data types)
2. Choose a suitable primary key (add one if no field is naturally unique)
3. Build in appropriate validation rules for fields (range, length, format, presence)
4. Read, complete and write basic SQL scripts

> ***Worked example – CUB scout database:*** Fields: CubNumber (primary key, text), CubName, DateOfBirth, Address, Gender (validated 'M' Or 'F', length 1, required), School, TelephoneNumber, DateJoined. Gender field shows all 3 validation types together: presence (Required = Yes), length (field size 1), format (validation rule 'M' Or 'F').

Key Terms

Database
structured collection of data for extracting info

Single-table database
a database containing only one table

Table
data about one type of item/person/event

Record
a row – all data about one entity

Field
a column – one item of data

Primary key
field with a unique value per record

Data type
classifies how data is stored & used

SQL
Structured Query Language – extracts data

SELECT / FROM
mandatory: fields to show / table to use

WHERE
filters records by a condition

ORDER BY
sorts the results

SUM / COUNT
total a field / count matching records

Practice Questions

Q1. Explain what is meant by a primary key and why FamilyName would not be a suitable choice for one. [3]

Q2. A database table BOOK has fields Title, Author, ISBN, Price, DateAdded. State a suitable data type for EACH of these five fields. [4]

Q3. Write an SQL query for the BOOK table that lists the Title and Author of all books with a Price greater than 10.00, sorted alphabetically by Author. [5]

Q4. Write an SQL query to count how many books in the BOOK table were written by 'J.K. Rowling'. [3]

Q5. Explain the difference between validation and verification in the context of a database, giving one example of each for a database field. [4]

Q6. A school wants a database of its teachers with these fields: TeacherID, FamilyName, Subject, YearsOfService, IsHeadOfDepartment, DateOfBirth. [6]

(a) State a suitable data type for each field. [3]

(b) Which field should be the primary key? Justify your answer. [2]

(c) Give one validation rule that could be applied to YearsOfService. [1]

Mark Scheme

A1. A primary key is a field whose value must be unique for every record, used to reliably identify a specific record (2). FamilyName is unsuitable because two or more people could share the same family name, so it would not be unique/could cause records to be confused (1).

A2. Title: text/alphanumeric (1); Author: text/alphanumeric (1); ISBN: text/alphanumeric (accept: could be the primary key) (1); Price: real (1); DateAdded: date/time (1) — award any 4 of 5 correct for full marks.

A3.

```
SELECT Title, Author
FROM BOOK
WHERE Price > 10.00
ORDER BY Author;
```

1 mark each for: SELECT with correct fields; FROM correct table; WHERE with correct field/operator/value; ORDER BY correct field; correct overall syntax/semicolon.

A4.

```
SELECT COUNT(Author)
FROM BOOK
WHERE Author = 'J.K. Rowling';
```

1 mark SELECT COUNT syntax; 1 mark correct FROM; 1 mark correct WHERE condition with quotes.

A5. Validation = an automated check that data is reasonable before it is accepted into the database (1), e.g. a range check on an age field rejecting a negative number (1). Verification = checking data has been copied accurately, e.g. via double entry of an email address / a screen check against a paper form (2).

A6. (a) TeacherID: text (1); FamilyName: text (1); Subject: text (1); YearsOfService: integer (1); IsHeadOfDepartment: Boolean (1); DateOfBirth: date/time (1) — award any 3 for the 3 marks. (b) TeacherID (1) because it is guaranteed to be unique for every teacher, unlike FamilyName which could repeat (1). (c) e.g. range check: value must be >= 0 (and reasonably <=50) (1).

— end of Chapter 9 —